

FHIA-04 — One History, Many Uses

REDUCING COSTS AND IMPROVING HEALTHCARE EFFECTIVENESS

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FUTURE HEALTHCARE INFORMATION ARCHITECTURE (FHIA) SERIES

A national longitudinal patient history can do far more than give providers better access to medical records. This paper explores how the same trusted information can support patients, providers, payers, researchers, manufacturers, and government—using secure, purpose-based access and AI to retrieve and analyze the information appropriate to each need. These capabilities could improve clinical decisions and patient outcomes, identify more effective treatments and providers, streamline prior authorization, expose ineffective or costly practices, accelerate research, and reduce healthcare costs.

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Introduction:

The previous paper in this series described how information distributed across healthcare organizations could be assembled into a trustworthy longitudinal patient history. That history connects the patient with providers, clinical events, physician notes, diagnoses and conditions, laboratory results, medical images, medications, treatments, outcomes, and payer information while preserving the original source evidence.

Why?

Imagine a healthcare information resource that could help:

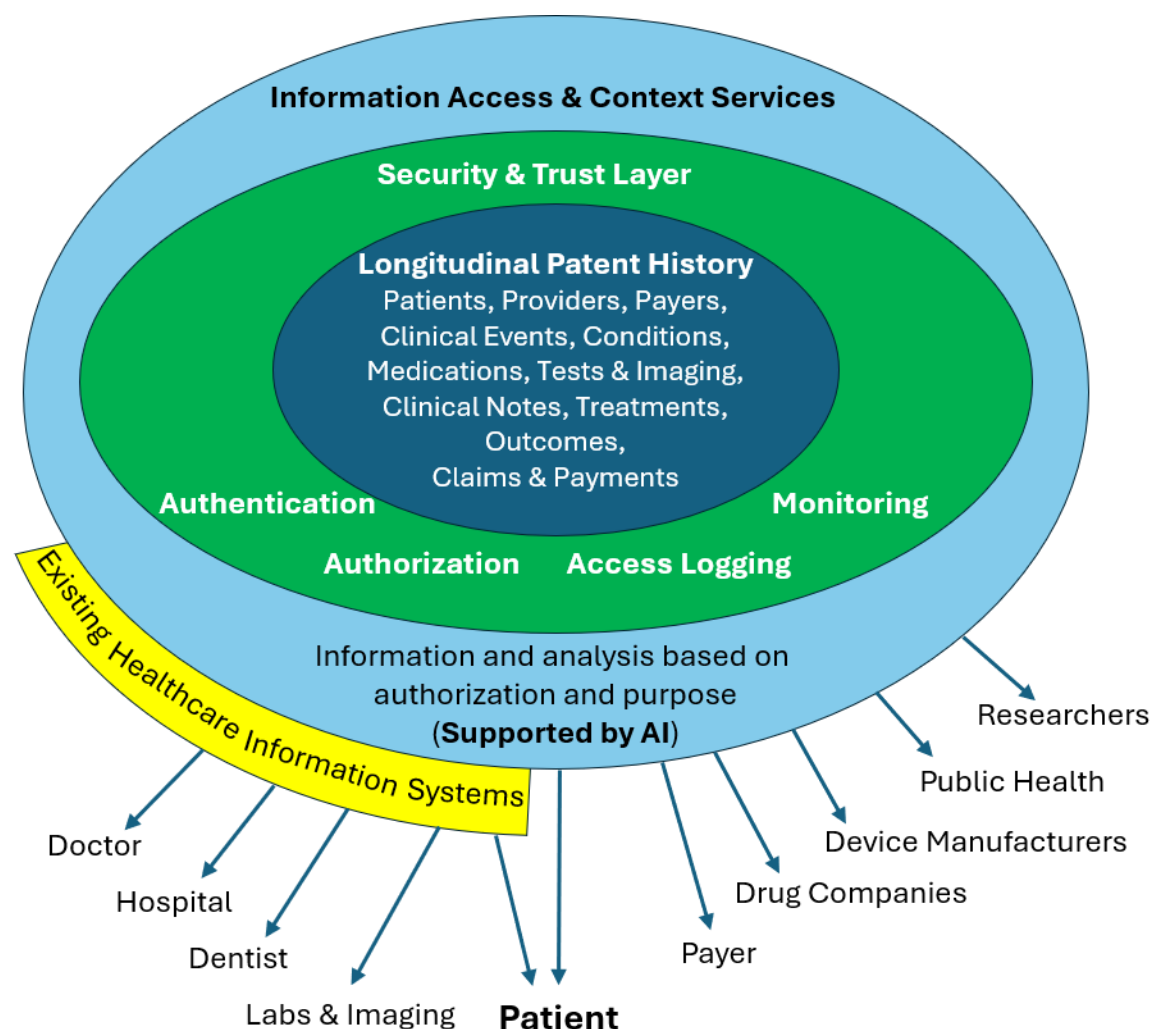
- **Physicians** immediately find the patient history relevant to the decision in front of them.
- **Patients** find providers with the most experience—and best outcomes—treating patients like them.
- **Families** identify patterns of disease across generations.
- **Hospitals and providers** learn which treatments produce the best outcomes for comparable patients.
- **Researchers** study disease progression, treatments, and outcomes across millions of longitudinal histories.
- **Drug companies** identify differences in effectiveness and adverse effects across patient populations.
- **Device manufacturers** track the long-term performance of implants and medical devices.
- **Payers** accelerate appropriate care, reduce unnecessary administrative effort, and learn when authorization policies increase rather than reduce total healthcare costs.
- **Patients and regulators** compare payer authorization and denial practices—and their effects on cost and patient outcomes.
- **Healthcare policymakers** understand where trillions of healthcare dollars are producing value—and where they are not.

These are not separate information problems requiring separate national databases. They are different uses of the same longitudinal healthcare information.

With U.S. healthcare spending measured in trillions of dollars annually, even small improvements in administrative efficiency, unnecessary care, treatment effectiveness,

adverse events, and provider selection can represent billions of dollars. The larger opportunity comes from addressing many of these opportunities simultaneously.

Realizing these benefits requires more than creating a longitudinal patient history. The architecture must make that information available for many different purposes while controlling who can access it, what they can access, and how it can be used.



One History Does Not Mean One Healthcare System

A national longitudinal patient history does not require a national healthcare delivery system, a national EHR, or federal control over healthcare decisions. Hospitals, physician practices, laboratories, pharmacies, dentists, rehabilitation facilities, payers, and other healthcare organizations would continue to operate their own Healthcare Information Systems and make their own clinical and business decisions.

The proposed architecture instead provides information capabilities that are difficult for individual healthcare organizations to provide independently. It creates a longitudinal

history that follows the patient across organizations and over time, while existing systems continue to support appointments, clinical documentation, orders, prescriptions, laboratory and imaging workflows, billing, claims, and the many other functions required to deliver and pay for healthcare.

This separation is important. No individual provider, payer, or technology vendor has a complete view of a patient's lifetime healthcare experience. Requiring every organization to independently locate, retrieve, reconcile, and interpret information from every other organization perpetuates duplication while still producing incomplete histories. A shared national information architecture performs these common functions once and makes the resulting information available to authorized users when needed.

The same principle applies to the broader uses described in this paper. Researchers, pharmaceutical companies, device manufacturers, payers, and policymakers do not need their own copies of a national patient database. They need controlled services that can answer authorized questions using appropriate portions of the longitudinal information.

The proposal centralizes information capabilities, not healthcare decisions. Existing healthcare organizations continue to deliver and manage healthcare. The national architecture provides a common information foundation that allows them—and other authorized users—to make better-informed decisions.

One History Does Not Mean Everyone Sees Everything

Bringing longitudinal healthcare information together does not mean making it universally accessible. The architecture must determine who is requesting information, why they are requesting it, and what information they are authorized to receive.

Patients would have access to their own identified longitudinal history. Healthcare providers directly involved in a patient's care could receive identified information appropriate to that care. Other users—such as researchers, pharmaceutical companies, device manufacturers, and healthcare analysts—could be authorized to use selected clinical and demographic information without receiving information that identifies individual patients.

In many cases, these users would not need access to individual records at all. A researcher might need to identify patterns among patients with particular characteristics. A pharmaceutical company might examine treatment effectiveness or adverse events across patient populations. An implant manufacturer might analyze device longevity and revision rates. The architecture could perform these analyses against the longitudinal information and return authorized results without providing copies of the underlying patient histories.

Security therefore involves more than protecting a database. Every request must be authenticated to establish who is making it, authorized to determine what that user is permitted to access, and constrained to information appropriate for the authorized purpose. Access should be logged so that there is an auditable record of who requested information, when it was requested, for what purpose, and what information was provided.

This creates an important distinction: **access to the national information resource does not necessarily mean access to patient records.** Different users can benefit from the same longitudinal information while receiving very different levels of detail.

One history can support many uses without giving every user the same view of that history.

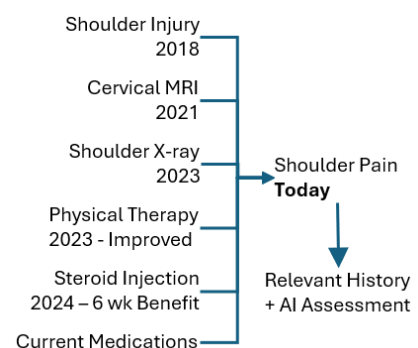
Different Questions Require Different Information

The value of a longitudinal patient history does not come from giving every authorized user access to everything it contains. It comes from identifying the information needed for a particular purpose and using that information to answer a question or support a decision.

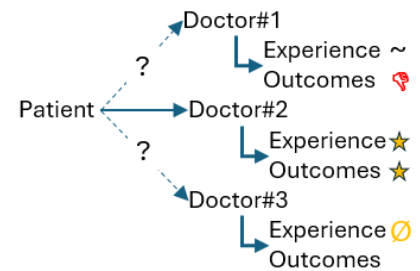
The Information Access & Context Services shown in the architecture provide this capability. After the Security & Trust Layer authenticates the user and determines what information that user is authorized to use, the services identify the information relevant to the request. AI can help understand the request, identify relevant relationships within the longitudinal history, analyze the authorized information, and present useful results. AI does not determine what information the user is authorized to access.

Consider several very different requests:

A physician evaluating a patient with recurring shoulder pain might need prior shoulder and neck conditions, injuries, imaging, treatments, medications, physical therapy, procedures, treatment outcomes, and recent clinical observations. AI could assemble the relevant history across years of care, identify changes and prior unsuccessful treatments, and summarize the findings with links to the supporting clinical evidence.



A patient recently diagnosed with a complex condition might ask which physicians have the most experience treating patients with similar characteristics, and which have achieved the best outcomes. The service could analyze de-identified histories of comparable patients, relate conditions to providers, treatments, and outcomes, and return comparative information without exposing the identities or histories of other patients.



A physician considering a treatment could ask how alternative treatments have been performed for patients similar to the one being treated. AI could identify an appropriate comparison population using relevant demographic and clinical characteristics, analyze treatments and outcomes, and present the results alongside information from the individual patient's history that may affect the decision.

A researcher studying disease progression might be authorized to analyze de-identified information for patients meeting defined clinical and demographic criteria. Rather than searching individual medical records, AI could help identify the appropriate cohort and analyze conditions, observations, laboratory results, medications, treatments, and outcomes across years or decades.

A pharmaceutical company investigating a potential adverse effect might need information about patients who received a particular medication, their relevant characteristics, dosage and duration, other medications, subsequent clinical events, laboratory results, and outcomes. The analysis could identify patterns and populations warranting further investigation without disclosing patient identities.

A medical-device manufacturer evaluating an implant might analyze device type, implantation date, relevant patient characteristics, subsequent procedures, complications, revisions, and outcomes. AI could identify patterns associated with device longevity or failure while the manufacturer receives only the information or aggregated results authorized for that purpose.

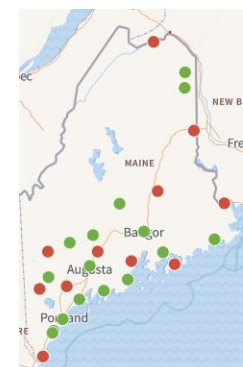
A payer evaluating a prior authorization request could use AI to assemble the relevant clinical evidence from the longitudinal history and determine whether the request satisfies established criteria, allowing appropriate requests to be approved quickly with little or no additional provider documentation. More complex requests could be evaluated using the patient's broader clinical history, previous treatments and outcomes, risk factors, the requesting provider's relevant experience, and outcomes for comparable patients. This is particularly important when a patient's history makes a routine authorization requirement inappropriate. Rather than simply reducing the cost of prior authorization, better

information could reduce delays in necessary care and identify situations in which denying or delaying a test or treatment ultimately produces worse outcomes and greater expense.

The longitudinal history could also **make payer performance measurable**. Authorization requests, approvals, denials, delays, appeals, subsequent treatment, patient outcomes, and total costs could be analyzed together to determine whether payer policies actually reduce unnecessary care or merely postpone necessary care and increase later costs. Aggregated results could identify unusually high denial rates, repeated requirements for treatments that prove ineffective, differences in approval times, and authorization practices associated with poorer outcomes. Such information could help regulators evaluate payer practices and eventually help employers and individuals consider payer performance when selecting health coverage.

A government agency investigating potential healthcare fraud could use claims, providers, patients, procedures, diagnoses, payments, clinical events, and other supporting information to identify unusual patterns that warrant investigation. AI could look for relationships and anomalies that are difficult to detect within a single payer's claims—for example, improbable procedure volumes, billing inconsistent with documented care, or patterns involving multiple providers and payers. Initial analysis could use de-identified information, with access to identified information requiring additional authorization when justified for an investigation.

Public-health planners evaluating access to healthcare could analyze the geographic distribution of patients, conditions, providers, specialties, services, and utilization to identify communities where available healthcare capacity does not match population needs. AI could identify areas with shortages of particular specialties, excessive travel for treatment, long delays in obtaining care, or growing demand associated with demographic and disease trends. The analysis could support decisions about where additional services are needed without exposing individual patient identities.



*Current/Former
birthing centers in
Maine*

These requests use the same longitudinal information resource, but they do not use the same information or expose the same level of detail.

Some require identified information about a specific patient. Others require de-identified information across populations. Still others may require only an aggregate result.

AI makes these capabilities practical because the relevant information may span millions of patients, decades of history, multiple types of clinical information, and complex relationships among conditions, treatments, providers, devices, and outcomes. The

architecture constrains AI to the information authorized for the user and purpose, while AI helps determine what within that authorized information is relevant and how to analyze it.

The objective is not to provide access to more data. It is to provide the right information, for the right purpose, to the right authorized user.

Conclusion — One History, Many Uses

The value of a longitudinal patient history extends far beyond giving a physician better access to a patient's medical records. The same information, securely accessed for different authorized purposes, can improve decisions throughout healthcare.

For patients, it can mean finding providers with the most relevant experience, understanding treatment alternatives, avoiding treatments that previously failed, and receiving care based on a more complete understanding of their history. For physicians, it can mean having the relevant information available when a decision must be made rather than searching through fragmented records or relying on what the patient can remember.

Across millions of patients, the same information becomes an extraordinarily valuable source of evidence. It can help determine which treatments work best for which patients, identify differences in provider experience and outcomes, detect drug and device problems earlier, expose patterns of fraud and unnecessary care, identify communities where healthcare services do not match population needs, and accelerate research that today depends on fragmented and incomplete information.

These benefits reinforce one another. Better information can lead to better treatment decisions. Measuring the results of those decisions can identify more effective treatments and providers. Applying that knowledge to future decisions can improve outcomes while avoiding ineffective care. Earlier detection of adverse drug effects, device failures, disease trends, and gaps in healthcare access can prevent problems rather than simply pay for their consequences.

The economic opportunity is equally broad. Savings do not have to come from a single dramatic change to healthcare. They can accumulate through less administrative work, fewer unnecessary tests and procedures, faster prior authorization, reduced fraud, fewer avoidable complications, better selection of treatments and providers, earlier identification of ineffective drugs and devices, and a better understanding of which healthcare expenditures actually improve outcomes. In a healthcare system spending trillions of dollars each year, improvements across many of these areas can have substantial cumulative value.

AI makes it practical to use this volume and complexity of longitudinal information, but AI is not the underlying asset. The asset is a trustworthy history connecting patients, conditions,

providers, treatments, costs, and outcomes across organizations and over time. AI helps retrieve, relate, analyze, and explain that information within the authorization and privacy constraints established by the architecture.

The ultimate measure of this architecture is therefore not how much healthcare data it collects or how sophisticated its AI becomes. It is whether people **live longer and healthier lives, receive more effective care from better-informed providers, avoid treatments that do not work, and spend less—individually and collectively—for the healthcare outcomes they achieve.**

One history. Many uses. Better decisions. Better outcomes. Lower cost.